

1. Write a program that prints the numbers from 10 down to 1 using a for loop, similar to a countdown timer.

```
for (let i = 10; i >= 1; i--) {  
    console.log(i);  
}
```

Output:

```
10  
9  
8  
7  
6  
5  
4  
3  
2  
1
```

**2. Create a menu-driven console app that lets the user: 1) View your favorite 3 IPL teams, 2) Add a new team, 3) Exit. Use a while loop to keep showing the menu until the user chooses Exit.

Hint: Use input() (or Scanner in Java) to get the user's choice each time.**

```
let teams = ["Mumbai Indians", "CSK", "RCB"];
```

```
let choice = 0;
```

```
while (choice != 3) {
```

```
    console.log("\n1. View IPL Teams");
```

```
    console.log("2. Add New Team");
```

```
    console.log("3. Exit");
```

```
choice = Number(prompt("Enter your choice:"));

if (choice == 1) {
    console.log("Favorite IPL Teams:");

    for (let i = 0; i < teams.length; i++) {
        console.log(teams[i]);
    }
}
else if (choice == 2) {
    let newTeam = prompt("Enter new team:");
    teams.push(newTeam);
    console.log("Team added successfully!");
}
else if (choice == 3) {
    console.log("Goodbye!");
}
else {
    console.log("Invalid choice!");
}
}
```

Menu:

1. View IPL Teams
 2. Add New Team
 3. Exit
-

**3. Build a 'Guess the Song' game like Spotify — the program randomly picks a song name from a list and asks the user to guess it. Use a do-while loop so the user can keep guessing until they get it right.

Constraint: Use at least 3 song names of your choice.**

```
let songs = ["Believer", "Shape of You", "Perfect"];

let randomIndex = Math.floor(Math.random() * songs.length);

let song = songs[randomIndex];

let guess;

do {
  guess = prompt("Guess the song name:");

  if (guess == song) {
    console.log("Correct! You guessed the song!");
  }
  else {
    console.log("Wrong guess! Try again.");
  }
} while (guess != song);
```

The program randomly selects **one song from 3 songs** and keeps asking until the user guesses correctly.

4. Explain with your own example the difference between entry-controlled and exit-controlled loops by writing a short code snippet for each (for/while vs do-while) and describing what happens if the loop condition is false at the start.

Entry-Controlled Loop — while

The condition is checked **before** the loop body runs.

```
let i = 1;
```

```
while (i <= 3) {  
  console.log(i);  
  i++;  
}
```

If the condition is **false at the beginning**, the loop runs **0 times**.

Example:

```
let i = 10;
```

```
while (i < 5) {  
  console.log(i);  
}
```

Output: Nothing, because $10 < 5$ is false.

Exit-Controlled Loop — do-while

The loop body runs **first**, and the condition is checked afterward.

```
let i = 1;
```

```
do {  
  console.log(i);  
  i++;  
} while (i <= 3);
```

If the condition is **false at the beginning**, the loop still runs **at least once**.

Example:

```
let i = 10;
```

```
do {
```

```
    console.log(i);  
} while (i < 5);
```

Output:

10

Easy Difference

Entry-Controlled	Exit-Controlled
for, while	do-while
Condition checked first	Condition checked after
May run 0 times	Runs at least 1 time